LESSON 10.6

Circumference and arc length

Date _____

Learning intention We are learning to find circumferences, arc lengths and perimeters of curved shapes.

Success criteria
- ☐ I can use $C = \pi d$ or $C = 2\pi r$ to find a circumference
- ☐ I can find an arc length: $\frac{\theta}{360} \times 2\pi r$
- ☐ I can find the perimeter of sectors and composite shapes

NOTES Copy what your teacher writes on the board.

Level 1 · Getting started

Easy · 12 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Find each circumference or perimeter, correct to one decimal place.

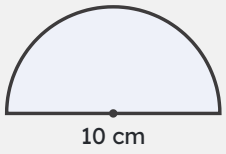
a Circumference



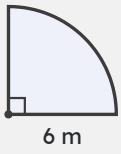
b Circumference



c Perimeter



d Perimeter



YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

12 QUESTIONS

REMEMBER $C = \pi d = 2\pi r$ · Semicircle $P = \frac{1}{2} \times \pi d + d$ · Quadrant $P = \frac{1}{4} \times 2\pi r + r + r$ · Use the π key.

ROUND 1 Questions 1–4: find the circumference of each circle with radius r, correct to one decimal place.

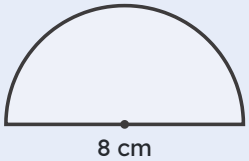
1. r = 3 cm
2. r = 7 m
3. r = 2.5 m
4. r = 1.5 km

ROUND 2 Questions 5–8: find the circumference of each circle with diameter d, correct to one decimal place.

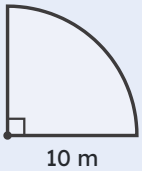
5. d = 8 cm
6. d = 15 m
7. d = 30 mm
8. d = 1.4 km

ROUND 3 Questions 9–12: find the perimeter of each shape, correct to one decimal place.

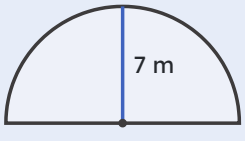
9.



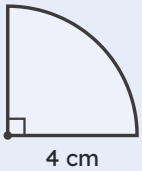
10.



11.



12.



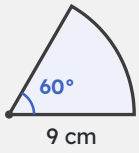
Level 2 · Building up

Medium · 16 questions

WE DO With your teacher: copy what they write on the board.

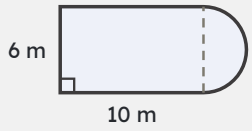
Example 2 **Give each answer correct to one decimal place.**

a Arc length



b Perimeter of the sector in part a

c Perimeter



d Radius of a circle with circumference 50 cm

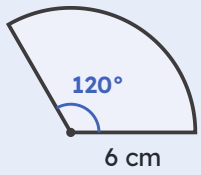
YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line.

16 QUESTIONS

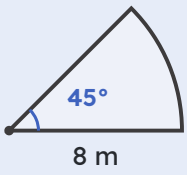
REMEMBER Arc length = $\frac{\theta}{360} \times 2\pi r$ (θ = the angle at the centre) · Sector P = arc + r + r · Composite: add the outside edges, not the dashed lines.

ROUND 1 Questions 1–4: find the perimeter of each sector, correct to one decimal place.

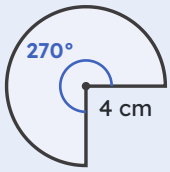
1



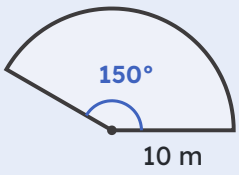
2



3



4

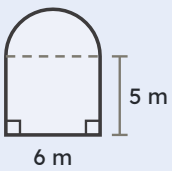


ROUND 2 Questions 5–8: find the perimeter of each shape, correct to one decimal place.

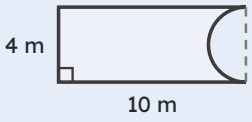
5



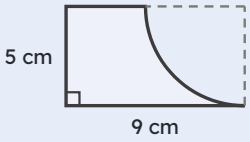
6



7 Cut-out: d = 4 m



8 Cut-out: r = 5 cm



ROUND 3 Questions 9–16: mixed. Read each one carefully: a tag shows an earlier lesson.

9 A wheel of diameter 64 cm turns 100 times. How far does it go, in m?

10 Find the perimeter of a rectangle 7.5 m by 4 m. 10.4

11 Convert 345 cm to m. 10.1

12 A round garden has a circumference of 18.85 m. Find its radius.

13 A kite has diagonals of 6 m and 5 m. Find its area. 10.5

14 Write the limits of accuracy of 4.5 m. 10.2

15 Find the arc length of a 120° sector with r = 3 m.

16 Find the percentage error of 20 cm, to the nearest cm. 10.3